

Lakshan Jayawardana

Undergraduate – BSc in Computer Science & Engineering

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Profile

Computer Science and Engineering undergraduate at the University of Moratuwa with hands-on experience in full-stack web development, embedded systems, digital design, and IoT projects. Proficient in programming, database management, and collaborative software development, with a strong interest in software engineering, cybersecurity, and emerging technologies.

Academic Qualifications

University of Moratuwa <i>Bachelor of Science in Computer Science and Engineering (2nd Year) GPA 3.70</i>	2024 – Present
Govt. Science College, Matale <i>GCE Advanced Level – 3As in Physical Science Stream Z-Score 2.4752</i> – GCE Ordinary Level – 9As	2019 – 2023
Ovilikanda Maha Vidyalyaya, Matale <i>Primary and Secondary Education</i>	2010 – 2018

Technical Skills

Programming Languages: Python, Java, C, C++
Web Development: HTML, CSS, JavaScript, React, Flask
Databases: MongoDB, MySQL
Hardware & HDL: VHDL, FPGA (BASYS 3), Digital Circuit Design
Tools & Platforms: Git, GitHub, VS Code, Vivado, Firebase

Projects

- UniMed: Voice-Enabled Digital Medical Record System** | Group Project – MORASHIFT 2026
– Developed a full-stack web application to digitize and manage student medical records, replacing a manual paper-based filing system.
– Implemented a Voice-to-Text interface that enables doctors to record diagnoses and prescriptions verbally, with manual correction capability.
– Designed secure storage and instant retrieval of medical records using student index numbers.
– **Technologies:** React, Flask REST API, MongoDB | **Team size:** 5
- Nanoprocessor Design – 4-bit Custom Processor** | Group Project – UoM 2026
– Designed and implemented a 4-bit nanoprocessor in VHDL capable of executing a custom instruction set including MOVI, ADD, NEG, and JZR.
– Designed the instruction decoder, register bank, ROM-based lookup components, and multiplexing logic for the processor architecture.
– **Technologies:** VHDL, Vivado, BASYS 3 FPGA | **Team size:** 4
- Smart Dustbin – IoT Waste Management System** | Group Project – UoM 2025
– Designed an IoT-based waste management system for public bus halts and industrial hotel/kitchen environments.
– Integrated ultrasonic, weight, and methane sensors with real-time notifications to monitor waste levels and odour conditions.
– Proposed solar-powered operation and a biogas recovery system to improve energy efficiency and reduce waste-management costs.
– **Technologies:** IoT Sensors, Solar Energy Systems, Mobile App Integration | **Team size:** 5

Work Experience

- Engineering Intern – Planning** 2024 (4 months)
DSP Controls (Pvt) Ltd, Sri Lanka (dspcontrols.lk)
– Supported engineering projects through project planning, design coordination, and preparation of technical ducting layout plans.
– Collaborated with engineering teams and gained practical exposure to technical documentation and project workflows.
- Trainee – Banking Operations** 2024 (6 months)
People's Bank, Matale Branch
– Assisted with daily banking operations, customer-facing transactions, financial record-keeping, and data management.
– Developed strong attention to detail, organizational skills, and professional communication in a structured work environment.

Achievements & Interests

- Served as a School Prefect (2020–2023), demonstrating leadership, responsibility, and organizational skills.
- Committee Member of the Astronomical Society, contributing to collaborative and academic activities.
- Strong interest in software engineering, cybersecurity, web development, and emerging technologies.

Languages

Sinhala: Native | **English:** Professional Working Proficiency